

MT 510: Colloids and Interfaces

Course Code	MT 510
Course Name	Colloids and Interfaces
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech./PG/Ph.D.
Mutual Exclusion	Some similarly with CY552: Hydrogen Generation and Storage
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction to colloid and interface:** colloids, classification of colloids based on affinity to carrier fluids, concept of stability of colloidal systems (DLVO and DLVO like theories and kinetics of coagulation and general principles of diffusion in a potential field/Brownian movement, physical characteristics of colloids, interfaces, Natures of interfaces, Thermodynamics of Interfaces, Adsorption and deposition at surfaces and interfaces (10 Hours)
- **Surface Tension, Adhesion and Capillarity:** Effect of confinement and finite size, concepts of surface and interfacial energies and tensions, Surface Tension, Contact Angle, Thermodynamics of Surfaces: Surface Tension as Surface Free Energy, Surface Tension: Implications for Curved Interfaces and Capillarity, Effects of Curved Interfaces on Phase Equilibria and Nucleation: The Kelvin Equation, Surface Tension and Contact Angle: Their Relation to Wetting and Spreading Phenomena, Contact Angles: Some Complications, Contact of Liquids with Porous Solids and Powders, Molecular Interpretation of Surface Tension, Apolar (Van Der Waals) and polar(acid-base) components of interfacial tensions. Young-Laplace equation of capillarity, examples of equilibrium surfaces, multiplicity. Free energies of adhesion, kinetics of capillary and confined flows (10 Hours)
- **The Rheology of Dispersions:** Introduction, Newton's Law of Viscosity, Concentric-Cylinder and Cone-and-Plate Viscometers, The Poiseuille Equation and Capillary Viscometers, The Equation of Motion: The Navier-Stokes Equation, Einstein's Theory of Viscosity of Dispersions, Deviations from the Einstein Model, Non-Newtonian Behavior, Viscosity of Polymer Solutions (10 Hours)
- **Van der Waals Forces:** Introduction and importance of Van der Waals Forces, Molecular Interactions and Power Laws, Molecular Origins and the Macroscopic Implications of van der Waals Forces, van der waals forces between large particles and over large Distances, calculating van der waals forces between macroscopic bodies, Theories of van der Waals Forces based on Bulk Properties, Effect of the medium on the van der Waals Attractions. (8 Hours)
- **Electric Double layer:** Background of surface charges and electrical, capacitor model of the double layer, diffuse double layer, Debye- Huckel Approximation, electrical double layer, Gouy-Chapman Theory, overlapping Double layer and interparticle repulsion, stern adsorption (8 Hours)

Laboratory Modules

1. NA

Text Books

1. P.C. Hiemenz and R. Rajagopalan (Editors), **Principles of Colloid and Surface Chemistry**, 3rd Edition, Academic Press, New York, 1997.
2. T. Cosgrove, **Colloid Science: Principles, Methods and Applications**, Wiley-Blackwell, 2005

References

1. NA